International Journal of Artificial Intelligence in Education (2025) 35:3922–3942
https://doi.org/10.1007/s40593-025-00520-7

ARTICLE

ChatGPT, Can You Make this Text Easier?: Investigating the Utility of LLMs in Easing Text Readability

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Received: 10 March 2025 / Accepted: 21 September 2025 / Published online: 27 October 2025

Abstract

Students' reading comprehension depends on their reading skill, the readability of the text, and the interaction of those two factors. For instance, developing readers benefit from easier texts – texts that use simpler words and are more cohesive. However, instructors may be unable to personalize texts for students due to time and knowledge constraints. Generative Large Language Models (LLMs) such as ChatGPT may afford instructors the ability to adapt texts to students' individual skill more easily and at a larger scale compared to manual editing. This study evaluates the ability of LLMs to ease text readability by assessing LLM-altered texts using natural language processing (NLP) techniques. In Experiment 1, 30 texts were rewritten by ChatGPT 4 to be more readable. In Experiment 2, the same 30 texts were rewritten by ChatGPT 4o and MagicSchool.ai to be readable for students at a particular grade. Both experiments used NLP tools to derive measures of linguistic sophistication, text cohesion, and holistic readability. Only the models and prompts tested in Experiment 2 demonstrated evidence of successfully easing text readability.

Keywords Text readability · Text comprehension · Natural language processing · Large language models

Introduction

Research in reading comprehension has found that students' reading comprehension can be supported by providing the students with skill-appropriate texts (McNamara & Kintsch, 1996). Students who are developing reading skills benefit from more readable texts and students with advanced reading skills benefit from less

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readable texts. (McNamara & Kintsch, 1996). This finding is difficult to implement in classroom settings for two reasons. The first is practical, instructors use texts to provide instruction and information on specific topics. For example, a physical science course uses text to introduce concepts to students (Herman & Wardrip, 2012). Thus, if all students in a classroom are to receive teaching on the same topic or concept, instructors must either find sets of texts with differing difficulties and the same content or manually edit a text to alter readability. Both options would consume teachers' already limited time (Teig et al., 2019). The second reason is that text readability is complex and depends on linguistic, syntactic, and discourse features of the text. The complexity of readability means that even well-meaning instructors may alter features of the text that are less essential to readability (Jensen et al., 2024). Though the importance of personalized text readability has been known for decades, there are few automated systems that rapidly edit texts to students' individual needs (McCarthy et al., 2025).

The advent of generative large language models (LLMs) such as ChatGPT may provide instructors with the ability to quickly alter text readability at scale. LLMs are capable of both generating texts on specific topics, as well as editing, summarizing, or otherwise changing texts that are uploaded within a graphical user interface (OpenAI, 2024). LLMs can generate and edit text quickly, fulfilling most queries within seconds. However, early reports on the efficacy of LLMs to alter text readability have mixed results. Researchers have found that LLMs are successful in rewriting a text to be easier (Butler et al., 2024; Li et al., 2023) but also null results when attempting to use LLMs to ease text readability (Imperial & Madabushi, 2023; Naseem, 2024). In addition, many of these studies rely on traditional reading formula such as the Flesch Kincaid Grade Level (FKGL: Kincaid et al., 1975), which computes readability as a function of syllables, words, and sentences in a text. More sophisticated measures of readability include features of language (e.g., word frequency) and syntax (e.g., word lemma repetition) and provide a holistic view of text readability (Crossley et al., 2017).

Further research on the use of LLMs to alter text readability is needed, and particularly, experiments that examine readability using a variety of measures of text readability. The current study responds to these needs by exploring the research question, "How can LLMs be used to ease text readability?". Two experiments utilized two LLM tools (ChatGPT, MagicSchool.ai) to alter text readability. The performance of the LLMs was assessed using traditional readability measures (e.g., Flesch-Kincaid Grade Level) and measures of linguistic and discourse features that assess text readability. more holistically.

Text Readability

Text readability can be broadly defined as the relative ease or difficulty with which readers can process or *comprehend* a text (Benjamin, 2012). Text readability includes word-level features of a text. For example, a text that uses common, easily decoded words is more readable (Benjamin, 2012; Crossley et al., 2017). In addition, readability encompasses discourse features of the text. For instance, a text that uses repetition of phrases to highlight a key idea is more readable (Crossley et al., 2017). Text readability and complexity is also dependent on the genre or purpose of the text. For example, texts in certain genres may have differing leels of readability because of the overall style of the genre. Research by Sheehan and colleagues (2014) on a corpus of more than 1,000 texts found that informational texts had fewer overall words and longer sentences compared to literary or narrative texts. In addition, the components of a text which influence readability may depend on the genre. For instance, the academic vocabulary in a text (e.g., the number of words in the text that commonly occur in textbooks) may be relatively low in a fiction novel that presents complex topics using simple words (Sheehan, 2016).

The study of text readability in the United States extends back to at least the late 1800 s and early 1900 s when researchers began to examine average sentence length between novels from different eras (Sherman, 1893), develop standardized reading tests (e.g., McCall & Crabbs, 1926), and testing adults' reading comprehension (Buswell, 1937). Over the 20th century, multiple formulas were proposed to capture the holistic readability of a text. In recent years, the Flesch-Kincaid Grade Level (FKGL) became the most widely used formula in educational practice (DuBay, 2004). As of the writing of this paper, a Google Scholar search for "Flesch Kincaid Grade Level" returns over 25,000 results, and the original report by Kincaid and colleagues has been cited over 4,400 times. The FKGL uses a simple formula that accounts for the number of syllables, words, and sentences in a text (see Fig. 1).

The FKGL, and its predecessor, the Flesch Reading-ease Test, inspired research into other readability measures (McClure, 1987) and were among more than 200 holistic readability measures created in the 20th century (DuBay, 2004). One readability measure that is still used in the 21 st century is the Dale-Chall readability test (DC: Chall & Dale, 1995). The DC is also a straightforward calculation like the FKGL. However, it does not include syllables but rather uses word lists to account for the number of "difficult words", in addition to total words and total sentences.

Over time, readability formulas such as FKGL and DC began to draw scrutiny as they were theorized to have weak construct validity. For example, research on reading comprehension repeatedly demonstrated the importance of discourse features for comprehension (Kintsch, 1988; McNamara & Kintsch, 1996; Ozuru et al., 2009). These traditional readability formulas do not account for lexical or discourse features that are central to theories of text comprehension (Kintsch, 1988; Koda, 2005). While readability formulas may broadly reflect text readability, more detailed measurements can provide more information about the holistic readability of a text. More recent work that utilizes natural language processing (NLP) has found that traditional formulas are less predictive of text readability and simplification when compared to more sophisticated measurements (Crossley et al., 2017, Tanprasert & Kauchak, 2021). For example, an analysis by Crossley and colleagues (2017) found that features of the text such as *lexical sophistication* and *text cohesion* better predicted adults' judgements of text readability and comprehension

$$0.39\left(\frac{\text{Total words}}{\text{Total sentences}}\right) + 11.8\left(\frac{\text{Total syllables}}{\text{Total words}}\right) - 15.59$$

Fig. 1 Flesch-Kincaid Grade Level Formula compared to traditional formulas. This finding aligns with work by Sheehan and colleagues (2014) who conducted a principal components analysis on linguistic features of a large text corpora, and found that several of the principal components were related to lexical sophistication (Academic Vocabulary, Concreteness, Word Unfamiliarity) and text cohesion (Degree of Narrativity, Cohesion).

The lexical sophistication of a text can be measured by assessing different features of the words of the text (Kyle et al., 2018; Sheehan et al., 2014). The sophistication of a single word can be measured as the frequency of the word in a text corpus, the age at which a native speaker acquires the word, the relative concreteness or abstractness of a word, or the average response time for an individual to recognize a word (Brysbaert et al., 2014; Kuperman et al., 2012; Kyle et al., 2018). Texts which contain less sophisticated words (e.g., words that are more frequent) are thought to be easier to read because the reader can more easily recall the meaning of the word during the reading process (Chen & Meurers, 2018).

The cohesion of a text is typically measured in two ways. First, by the overlap in word lemma between sentences and paragraphs, particularly the overlap in content words across sentences. (Sheehan et al., 2014). A word lemma is the dictionary form of a word that represents all other possible forms. For example, "soft" represents all words in the list, "softly, softer, softest". Texts that use more overlapping word lemma are theorized to be easier to read because readers can leverage the overlapping lemma to better connect different sentences and paragraphs (Crossley et al., 2017; McNamara & Kintsch, 1996). Second, text cohesion can be measured by the number of connectives (e.g., as a result, because, therefore) used in the text. More connectives in the text afford better readability as connectives more clearly relate different ideas to each other (Cain & Nash, 2011).

Overall, a more holistic picture of text readability emerges by utilizing measures of lexical sophistication and text cohesion along with readability formulas. In the following experiments, the ability of LLMs to alter text readability is assessed using all three types of measures. The use of multidimensional measures allows investigation into the types of text alterations that LLMs can perform readily, and alterations which require more human interaction with LLMs. While there are other metrics that could be used to examine holistic text readability, the focus of this paper is on lexical sophistication and text cohesion due to repeated findings in the reading sciences that those two features of a text predict comprehension (Crossley et al., 2017; McNamara & Kintsch, 1996).

Generative LLMs and Text Readability

The use of generative large language models (LLMs) in education settings has become increasingly common since the public release of ChatGPT in the fall of 2022 (Moore et al., 2023). The rise in LLM usage includes general-use LLM tools (e.g., ChatGPT 4, Gemini) and LLM tools specifically adapted for education (e.g., MagicSchool.ai, MerlynMind). Users can interact with LLMs via a chat-based interface and upload common file formats for the LLM to use in response-generation.

The explosion of LLM use in education and learning includes early efforts to ease text readability. Many of these efforts have been focused on simplifying medical texts for lay readers. For instance, Butler and colleagues (2024) and Li and colleagues (2023) both used LLMs to simplify medical handouts and patient reports. In both studies, the researchers reported that the LLM-generated texts had significantly lower FKGL scores compared to the original texts (Butler et al., 2024; Li et al., 2023). In contrast, Naseem (2024) and Imperial and Madabushi (2023) both found that LLMs could not reliably produce texts levelled to appropriate grades.

One limitation of the current directions in LLM text readability research is that studies have utilized traditional readability formulas to assess readability. FKGL provides a broad overview of readability by assessing word and sentence length. However, more sophisticated measurements of the linguistic and syntactic features of a text are better predictors of text readability (Crossley et al., 2017, Tanprasert & Kauchak, 2021). Indeed, Imperial and Madabushi (2023) noted this limitation in their paper, writing, "We are well aware of the limitations of FKGL for evaluating the performances of simplification systems as highlighted in Tanprasert and Kauchak." (pp.9). Thus, further assessment of the efficacy of LLMs to alter text readability must consider more varied measures.

The Current Study

The current study tests the ability of generative LLMs to simplify texts by assessing LLM-generated texts using NLP techniques. Two experiments were conducted, the first on ChatGPT 4 (OpenAI, 2023), the second on ChatGPT 4o (OpenAI, 2024) and MagicSchool.ai (Foster et al., 2024). In both experiments, the LLM tools altered a set of science texts, comparing the readability of the generated texts to the readability of the original texts. The texts were all informational, which potentially afforded greater opportunities for the LLM tools to simplify the texts by containing longer sentences and more sophisticated words compared to narrative texts (Sheehan et al., 2014). However, this decision limits the generalizability of the study due to noted differences between informational and narrative texts.

The prompts utilized were simple one sentence prompts (see Table 1). There were two reasons for using simple prompts. First, these types of straightforward prompts may be similar to those used in current instructional practice. Recent reports on the use of Generative AI in education have found fewer than 15% of teachers report using

Table 1 ChatGPT prompts to rewrite the texts

Prompt number	Prompt text
Experiment 1, Prompt 1	Rewrite this text in plain language.
Experiment 1, Prompt 2	Rewrite this text to be easier on the Flesch Kincaid Grade Level test.
Experiment 1, Prompt 3	Rewrite this with more connectives and explanations.
Experiment 2 Prompt	Rewrite this text for a 6th grade student.

generative AI for their educational practice (Pantazatos et al., 2024). In addition, there is limited guidance on how to best use LLMs for text readability tasks and competing demands on teachers' time may prevent them from using prompt engineering techniques. Second, using simple prompts provides a baseline for LLM performance in this type of task, which can be used in future studies that leverage more complex LLM interactions (e.g., Prompt Engineering, Retrieval-Augmented Generation) to alter text readability can be compared to this baseline performance.

The use of simple prompts may limit the full capabilities of a generative LLMs. However, past research on using LLMs to represent information from texts found diminishing returns when including more detailed prompts and methodologies (e.g., chain of thought reasoning; Polat et al., 2025). Thus, empirically testing the abilities of generative LLMs to alter text cohesion with limited information can both fully test the capabilities of generative LLMs in sparse information tasks, as well as provide a reflection of current uses of AI in pedagogical practice.

The experiments were preregistered on the Open Science Framework[1]. The original preregistration was for only Experiment 1 and was updated to include Experiment 2 once the decision was made to conduct a follow-up experiment. In addition, the statistical models were changed from the planned ANOVAs to Multivariate GLM models because multivariate GLM models control for multiple dependent variables – particularly dependent variables that are correlated (Johnson, 2023). The text corpus and data for both experiments are available for peer-review at the open science framework via an anonymized link[2]. The anonymized link will be updated to the public DOI upon acceptance of this paper for publication.

Experiment 1

Experiment 1 was conducted in March of 2024 and assessed the performance of ChatGPT 4 in easing text readability using simple prompts. ChatGPT 4 was tasked with modifying texts to improve their readability using three different prompts (see Table 1). The original and modified texts were assessed using measures of linguistic sophistication, text cohesion, and traditional readability formulas. Three hypotheses were tested to discern if ChatGPT 4 effectively ease text readability.

H1.1: The modified texts would have lower lexical sophistication scores compared to the original texts (i.e., the modified texts would use simpler words).
H1.2: The modified texts would have higher text cohesion scores compared to the original texts (i.e., the modified texts would be more cohesive).
H1.3: The modified texts would have lower readability grade level scores compared to the original texts (i.e., the modified texts could be read by students in lower grades).

[1] https://doi.org/10.17605/OSF.IO/XZPBJ.

[2] https://osf.io/m7qvf/?view_only=53a39b93172244dfaa1a4f75bb257b25.

Experiment 1 Method

Texts

A set of 30 science texts were selected for use in the experiment. The texts were modified by ChatGPT 4 using three different prompts (see Table 1). Thus, each text had four versions (Text Version: Original, Prompt 1, Prompt 2, Prompt 3) resulting in a total of 120 texts.

The texts were drawn from a larger set of texts used in the iSTART reading software (Snow, 2016) and chosen for two reasons. First, the texts were rated by experts for grade readability, and the levels ranged from 8th grade to 12th grade (Snow, 2016). Second, the texts discussed a range of topics in biology and chemistry that are listed on the CommonCore standards for 6th grade, which students are expected to have received instruction about by 8th grade. While the current study does not include human-subjects validation, the topic selection affords future studies with middle school, high school, and college students using the same text corpus.

Natural Language Processing Indices

For each of the 120 texts, four indices were derived to measure lexical sophistication, four indices were derived to measure text cohesion, and two traditional measures of readability were derived (see Table 2). The selected indices were chosen based on past research on the correlation between NLP indices and text readability (e.g., Crossley et al., 2017). In addition, the word count of each text was measured to use as a covariate, as the derived NLP indices can be affected by the total number of words in the text.

The NLP indices were derived using existing NLP tools. The Tool for the Automatic Assessment of Lexical Sophistication (TAALES: Kyle et al., 2018) was used to derive the four indices of lexical sophistication. The Tool for the Automatic Assessment of Cohesion (TAACO: Crossley et al., 2019) was used to derive the four measures of text cohesion. Finally, the Automatic Readability Tool for English (ARTE: Choi & Crossley, 2022) was used to derive the traditional readability measures.

Six of the selected measures (Frequency, Age of Acquisition, Concreteness, Lexical Decision Response Time, Sentence Lemma Overlap and Paragraph Lemma Overlap) can be derived either from all of the words in the text, or only the content words (i.e., nouns, verbs, adverbs, adjectives) in the text. Both measures were derived, and subsequent analyses were tested using both the all-word derivations and the content word derivations. The results of the experiment did not differ based on the derivation used, for that reason only the results of the all-word derivations are reported in the results section.

Table 2 Natural language processing indices

Index title	Index measurement	Index construct
Word Frequency	Each word in a text is assigned a frequency score (higher = more frequent) using the SUBTLEXus Corpus. Consistent with SUBTLEX reporting guidelines, the reported score is frequency per million words. The average score for the text is reported.	Lexical Sophistication
Word Age of Acquisition	Each word in a text is assigned an Age of Acquisition score (lower = learned at a younger age) using the Kuperman Corpus. The average score for the text is reported.	Lexical Sophistication
Word Concreteness	Each word in a text is assigned a Concreteness score (higher = more concrete) using the Brysbaert Corpus. The average score for the text is reported.	Lexical Sophistication
Lexical Decision Response Time	Each word in a text is assigned a lexical decision response time measured in milliseconds (lower = faster response time) using the English Lexicon Project Corpus. The average score for all the words in the text is reported.	Lexical Sophistication
Lemma Overlap in Adjacent Sentences	Each sentence (except the last sentence) is assigned a score using the formula: number of lemma that occur in the following sentence divided by the total number of lemma in the sentence. The average score for all sentences is reported.	Text Cohesion
Lemma Overlap in Adjacent Paragraphs	Each paragraph (except the last paragraph) is assigned a score using the formula: number of lemma that occur in the following paragraph divided by the total number of lemma in the paragraph. The average score for all paragraphs is reported.	Text Cohesion
All Causal Connectives	The proportion of words in the text that are causal connectives (e.g., arises, because, therefore).	Text Cohesion
All Connectives	The proportion of words in the text that are connectives (e.g., actually, a consequence, nonetheless)	Text Cohesion
Dale-Chall Readability	The text score using the Dale-Chall readability test.	Readability Formula
Flesch-Kincaid Readability	The text score using the Flesch-Kincaid readability test.	Readability Formula

Experiment 1 Results

The descriptive statistics of the NLP indices were examined by text version (Original, Prompt 1, Prompt 2, Prompt 3) to identify potential changes based on text version. Table 3 shows the descriptive statistics by group. There was a statistically significant difference in word count among the original and edited texts. The text produced by ChatGPT had fewer words on average than the original texts. Figure 2 shows the differences by text and prompt. For each individual original text, the three altered texts had differences in word count based on the prompt condition. In addition, between prompts there were differences in whether the altered texts had slightly or significantly fewer words compared to the original texts. However, the overall trend was negative, and thus required further analyses to covary word count, as word count affects some of the NLP indices selected for analyses (e.g., a longer text is likely to have more repeated lemmas).

One potential confound was due to the nature of the texts and the corpora used for analyses (e.g., word frequency based on the SUBTLEX corpus). The texts were expository science texts, and the corpora were created from general English texts

Table 3 Experiment 1: text corpus coverage

Index	Original text		Prompt 1		Prompt 2		Prompt 3	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Word Count	435.20	44.03	285.83	30.36	275.60	24.96	434.27	30.46
Word Frequency	292727.75	41034.57	241619.42	49080.52	242762.93	49714.52	252842.45	36587.81
Word Age of Acquisition	6.10	0.38	6.20	0.42	6.03	0.48	6.61	0.30
Word Concreteness	2.58	0.14	2.60	0.15	2.64	0.13	2.53	0.12
Lexical Decision Response Time	640.25	7.65	640.97	7.34	638.60	7.36	650.56	5.51
Lemma Overlap in Adjacent Sentences	0.23	0.05	0.17	0.04	0.18	0.04	0.20	0.04
Lemma Overlap in Adjacent Paragraphs	0.32	0.06	0.26	0.05	0.27	0.05	0.26	0.04
All Causal Connectives	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01
All Connectives	0.06	0.01	0.07	0.02	0.08	0.02	0.07	0.01
Dale-Chall Readability	9.51	1.08	9.91	1.05	9.41	1.13	11.12	0.79
Flesch-Kincaid Readability	9.78	1.99	10.94	1.59	9.78	1.74	14.14	1.35

and experiments. Thus, any rewriting of the original texts may artificially inflate or deflate the corpora-based measurements if the words in the either group of texts were more or less likely to appear in the corpora. Thus, for each text, the proportion of words in the text that occur in the corpora was calculated, and the means and standard deviations are presented in Table 4.

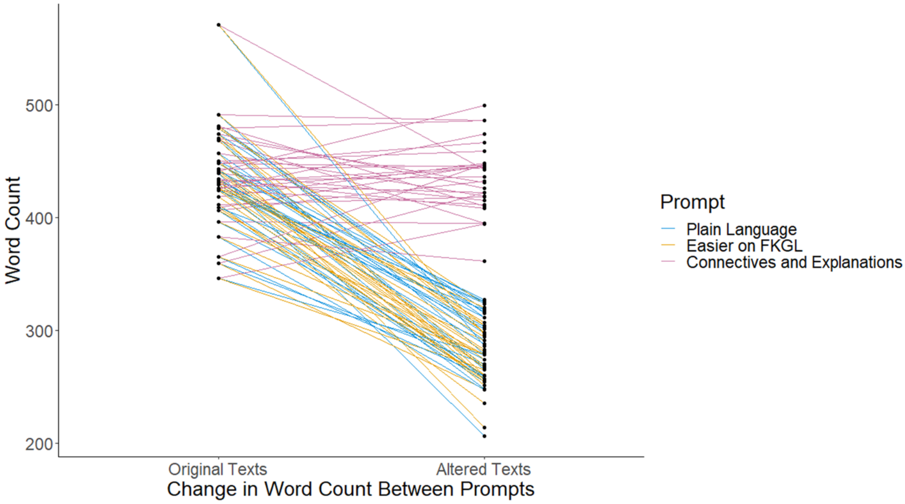

Fig. 2 Experiment 1: word count by text and prompt

Table 4 Experiment 1: Text corpus coverage

Corpora	Original text		Prompt 1		Prompt 2		Prompt 3	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
SUBTLEX	0.956	0.030	0.959	0.022	0.958	0.027	0.957	0.024
Brysbaert Concreteness	0.813	0.046	0.797	0.045	0.802	0.053	0.809	0.046
Age of Acquisition	0.722	0.042	0.710	0.043	0.713	0.048	0.723	0.035
English Lexicon Project	0.954	0.032	0.959	0.026	0.966	0.030	0.959	0.031

Table 5 Experiment 1: Multivariate GLM predicting NLP indices as a function of text Version, holding word count constant

Dependent Variable	F	*P*
Word Frequency	9.588	< 0.001
Word Age of Acquisition	11.838	< 0.001
Word Concreteness	3.620	0.008
Lexical Decision Response Time	16.303	< 0.001
Lemma Overlap in Adjacent Sentences	8.008	< 0.001
Lemma Overlap in Adjacent Paragraphs	8.531	< 0.001
All Causal Connectives	1.020	0.400
All Connectives	2.076	0.088
Dale-Chall Readability	15.409	< 0.001
Flesch-Kincaid Readability	33.364	< 0.001

Hypothesis Testing

In order to test the three hypotheses, a multivariate general linear model was constructed with the ten NLP indices as the dependent variables, the text version (Original, Prompt 1, Prompt 2, Prompt 3) as the fixed factor, and word count as a covariate. Table 5 shows the F and p values for all the DVs. The results of the model showed that there were significant differences as a function of prompt for eight of the ten DVs (all excepting Causal Connectives and All Connectives).

The model produced estimated marginal means (EMMs) for each the dependent variables on all the text versions, holding word count constant. The EMMs were assessed using Bonferroni post-hoc tests, and the results of the post-hoc tests were used to evaluate the three hypotheses. Table 6 shows the EMMs and the results of the post-hoc tests for the eight significant models (i.e., not including the models for All Connectives and Causal Connectives).

Table 6 Experiment 1: Estimated marginal means and Post-hoc test results

Dependent variable (Construct)	Text version	Estimated marginal mean	Std. Error	Mean difference from original text	*p*
Word Frequency (Lexical Sophistication)	Original	263384.96	12130.05		
	Prompt 1	268847.59	11626.94	−5462.64	1.00
	Prompt 2	273866.86	12560.83	−10481.90	1.00
	Prompt 3	223853.14	12044.83	39531.81	**0.03**
Age of Acquisition (Lexical Sophistication)	Original	6.32	0.11		
	Prompt 1	6.00	0.11	0.32	0.61
	Prompt 2	5.80	0.11	0.52	0.07
	Prompt 3	6.83	0.11	−0.51	**0.01**
Word Concreteness (Lexical Sophistication)	Original	2.53	0.04		
	Prompt 1	2.65	0.04	−0.13	0.34
	Prompt 2	2.69	0.04	−0.17	0.10
	Prompt 3	2.48	0.04	0.05	0.88
Lexical Decision Response Time (Lexical Sophistication)	Original	644.52	1.93		
	Prompt 1	637.01	1.85	7.51	0.16
	Prompt 2	634.07	2.00	10.45	**0.02**
	Prompt 3	654.78	1.91	−10.26	**0.01**
Lemma Overlap in Adjacent Sentences (Text Cohesion)	Original	0.22	0.01		
	Prompt 1	0.18	0.01	0.04	0.42
	Prompt 2	0.19	0.01	0.03	1.00
	Prompt 3	0.19	0.01	0.03	0.07
Lemma Overlap in Adjacent Paragraphs (Text Cohesion)	Original	0.33	0.01		
	Prompt 1	0.26	0.01	0.07	**0.04**
	Prompt 2	0.26	0.02	0.06	0.12
	Prompt 3	0.26	0.01	0.06	**0.01**
Dale-Chall Readability (Readability Formula)	Original	10.04	0.28		
	Prompt 1	9.43	0.27	0.61	1.00
	Prompt 2	8.86	0.29	1.18	0.14
	Prompt 3	11.64	0.28	−1.60	**0.01**
Flesch-Kincaid Grade Level (Readability Formula)	Original	9.87	0.48		
	Prompt 1	10.86	0.46	−0.99	1.00
	Prompt 2	9.69	0.50	0.18	1.00
	Prompt 3	14.22	0.48	−4.35	**0.01**

Note. The EMMs are evaluated at Word Count = 357.72. **Bolded** p-values are significant

Overall, the three hypotheses were not supported, as none of the three altered text versions (Prompt 1, Prompt 2, Prompt 3) showed consistent evidence of lower lexical sophistication, higher text cohesion, or easier readability scores.

Close examination of the EMMs and post-hoc tests, however, showed that one of the text versions (Prompt 3) was significantly less readable than the original texts. The Prompt 3 text version used significantly more sophisticated words and was rated as more difficult by both of the traditional readability formulae.

Experiment 1 Discussion

The goal of Experiment 1 was to assess the performance of ChatGPT 4 in easing text readability using simple prompts. Three different prompts were used for ChatGPT 4 to modify texts to improve their readability. The resulting texts were assessed using readability formulas, measures of linguistic sophistication, and measures of text cohesion. It was hypothesized that ChatGPT 4 would improve readability as measured by NLP indices of lexical sophistication, text cohesion, and traditional readability formulas. The results of the experiment do not support any of the hypotheses.

Experiment 1 had two additional noteworthy findings. First, the texts altered by ChatGPT had significantly fewer words compared to the original texts. The difference was most pronounced in the texts produced by Prompts 1 and 2 (see Table 1 for the prompts). Second, the texts produced by Prompt 3 were in fact less readable than the original texts, despite the prompt specifying the types of language (connectives and explanations) typically associated with more readable texts.

Overall, Experiment 1 demonstrated that operationalizing LLMs using simple prompts is unlikely to produce more readable texts. There are two potential pathways to improve the performance of generative LLMs for the task of easing text readability. First, different prompts have been shown to improve the performance of generative LLMs on text-based tasks (Pawlik, 2025). In addition, new generations of publicly available generative LLMs have shown improvements in task performance over previous generations (OpenAI, 2024).

Experiment 2

Experiment 2 was conducted in June of 2024 and extended the design and procedure of Experiment 1 by using a different prompt for ChatGPT (see Table 1) and using an LLM-based tool specifically designed to alter text readability (MagicSchool.ai Text Leveler). In addition, the timing of this Experiment (i.e., approximately three months after Experiment 1) allowed the use of a newer public version of ChatGPT compared to Experiment 1 (ChatGPT 4o vs. ChatGPT 4).

Both ChatGPT 4o and MagicSchool.ai were prompted to modify texts to improve their readability. ChatGPT 4o was given a natural language prompt to alter a text to a specific grade level, and the MagicSchool.ai text leveler tool (see Fig. 2) was used to specify a grade level (6th grade for both tools). The original and modified texts were assessed using the same measures of readability as Experiment 1: Linguistic sophistication, text cohesion, readability formulas. Three hypotheses were tested to discern if ChatGPT 4o and MagicSchool.ai effectively improved text readability.

> H2.1: The modified texts would have lower lexical sophistication scores compared to the original texts (i.e., the modified texts would use simpler words).
> H2.2: The modified texts would have higher text cohesion scores compared to the original texts (i.e., the modified texts would be more cohesive).
> H2.3: The modified texts would have lower readability grade level scores compared to the original texts (i.e., the modified texts could be read by students in lower grades).

Experiment 2 Method

Texts

The same set of 30 science (e.g., Chemistry, Biology) texts from Experiment 1 were modified by ChatGPT 4o and by the MagicSchool.ai Text Leveler Tool (see Fig. 3). ChatGPT 4o was prompted to, "Rewrite this text for a 6th grade student" (see Table 1). The MagicSchool.ai Text Leveler tool was set to rewrite the text for a 6th grade student. Thus, each text had three versions (Original, ChatGPT, MagicSchool.ai) resulting in a total of 90 texts. 6th grade was chosen as the target because the original text corpus contained texts for grades 8–12, meaning that levelling the texts to a 6th grade reading level required easing readability for all texts (see Experiment 1 Method section).

Natural Language Processing Indices

The same NLP indices derived in Experiment 1 were derived in Experiment 2 (see Table 2 for the index descriptions). Word count was also measured to be used as a covariate. In addition, the same procedure was used to test the NLP indices that could be derived from all the words in the text or the content words in the text. The same pattern of results was found using either derivation and the following results are reported using the all-words derivations.

Experiment 2 Results

The descriptive statistics of the NLP indices were examined by text version (Original, ChatGPT, MagicSchool.ai) to identify potential changes based on text version. Table 7 shows the descriptive statistics by group. There was a difference in word count between the original and edited texts. The texts produced by both ChatGPT and MagicSchool.ai had fewer words on average than the original texts. Figure 4 shows the differences by text and prompt. This difference required further analyses to covary word count. In addition, the same test of text corpus coverage as in Experiment 1 was conducted to ensure that the corpus-based measurements were not confounded, and the results are shown in Table 8.

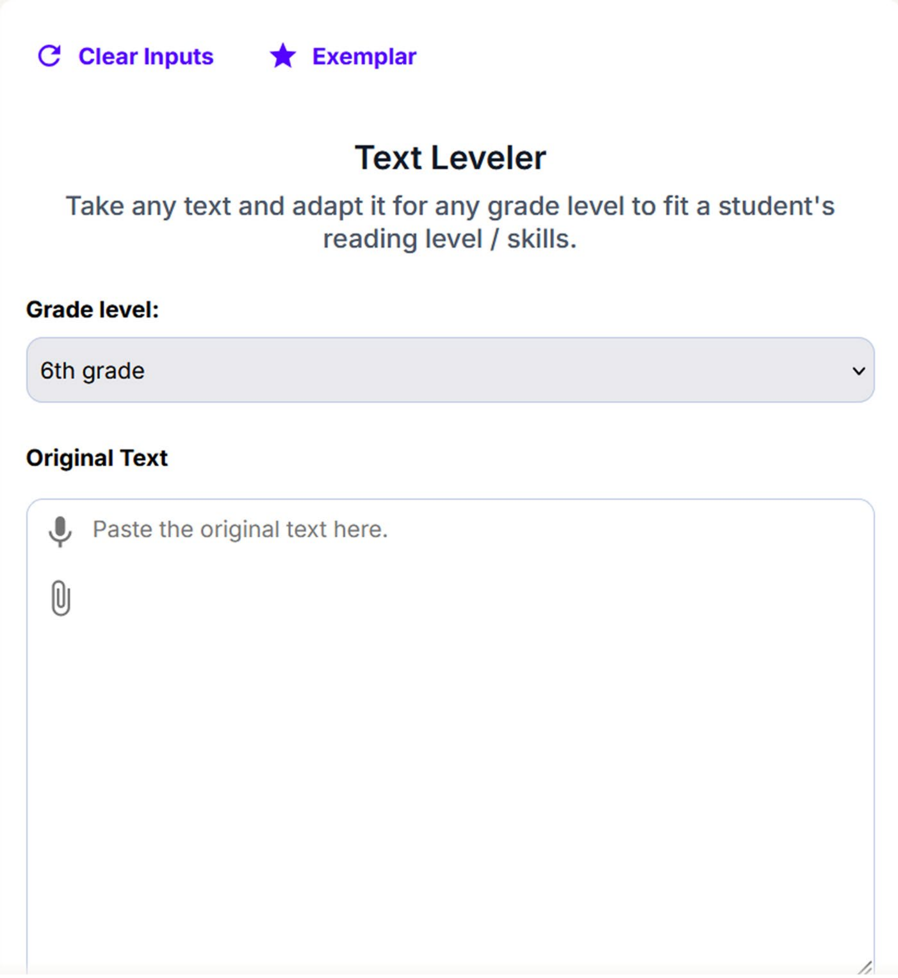

Fig. 3 MagicSchool.ai Text Leveler Tool

Hypothesis Testing

In order to test the three hypotheses, a multivariate general linear model was constructed with the ten NLP indices as the dependent variables, the text version (Original, ChatGPT, MagicSchool.ai) as the fixed factor, and word count as a covariate. Table 9 shows the F and p values for all the DVs. The results of the model showed that there were significant differences as a function of prompt for eight of the ten DVs (all excepting Causal Connectives and All Connectives).

The model produced estimated marginal means (EMMs) for each the dependent variables on all the text versions, holding word count constant. The EMMs were assessed using Bonferroni post-hoc tests, and the results of the post-hoc tests were used to evaluate the three hypotheses. Table 10 shows the EMMs and the results of

Table 7 Experiment 2: Descriptive statistics

Index	Original text		ChatGPT		MagicSchool.ai	
	Mean	SD	Mean	SD	Mean	SD
Word Count	435.20	44.03	337.20	37.04	397.07	41.87
Word Frequency	292727.75	41034.57	241619.42	49080.52	262886.00	40225.45
Word Age of Acquisition	6.10	0.38	5.77	0.36	5.78	0.36
Word Concreteness	2.58	0.14	2.68	0.15	2.65	0.14
Lexical Decision Response Time	640.25	7.65	633.14	5.56	633.07	6.21
Lemma Overlap in Adjacent Sentences	0.23	0.05	0.18	0.04	0.19	0.04
Lemma Overlap in Adjacent Paragraphs	0.32	0.06	0.26	0.05	0.31	0.05
All Causal Connectives	0.01	0.01	0.02	0.01	0.01	0.01
All Connectives	0.06	0.01	0.07	0.02	0.07	0.02
Dale-Chall Readability	9.51	1.08	8.77	0.93	8.62	0.95
Flesch-Kincaid Readability	9.78	1.99	8.16	1.37	8.30	1.36

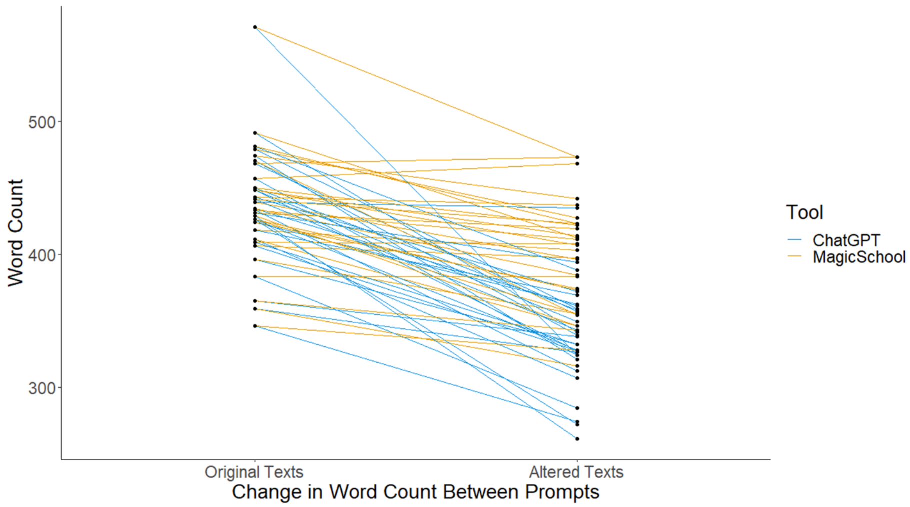

Fig. 4 Experiment 2: Word count by text and prompt

Table 8 Experiment 2: Text corpus coverage

Corpora	Original text		ChatGPT		MagicSchool.ai	
	Mean	SD	Mean	SD	Mean	SD
SUBTLEX	0.956	0.030	0.952	0.031	0.961	0.031
Brysbaert Concreteness	0.813	0.046	0.794	0.044	0.810	0.046
Age of Acquisition	0.722	0.042	0.712	0.047	0.723	0.046
English Lexicon Project	0.954	0.032	0.957	0.030	0.961	0.032

Table 9 Experiment 2: Multivariate GLM predicting NLP indices as a function of text Version, holding word count constant

Dependent variable	F	P
Word Frequency	3.757	0.014
Word Age of Acquisition	7.633	< 0.001
Word Concreteness	5.484	0.002
Lexical Decision Response Time	9.174	< 0.001
Lemma Overlap in Adjacent Sentences	8.661	< 0.001
Lemma Overlap in Adjacent Paragraphs	9.203	< 0.001
All Causal Connectives	1.769	0.159
All Connectives	1.092	0.357
Dale-Chall Readability	6.455	< 0.001
Flesch-Kincaid Readability	7.146	< 0.001

Table 10 Experiment 2: estimated marginal means and Post-hoc test results

Dependent variable (Construct	Text version	Estimated marginal mean	Std. error	Mean difference from original text	p
Word Frequency (Lexical Sophistication)	Original	295505.28	9965.53		
	ChatGPT	252533.99	10461.89	42971.29	0.04
	MagicSchool.ai	263329.46	8405.54	32175.83	0.04
Age of Acquisition (Lexical Sophistication)	Original	6.21	0.08		
	ChatGPT	5.65	0.08	0.56	**0.01**
	MagicSchool.ai	5.80	0.07	0.41	**0.01**
Word Concreteness (Lexical Sophistication)	Original	2.53	0.03		
	ChatGPT	2.73	0.03	−0.20	**0.01**
	MagicSchool.ai	2.64	0.03	−0.11	**0.01**
Lexical Decision Response Time (Lexical Sophistication)	Original	641.59	1.41		
	ChatGPT	631.59	1.48	9.99	**0.01**
	MagicSchool.ai	633.28	1.19	8.30	**0.01**
Lemma Overlap in Adjacent Sentences (Text Cohesion)	Original	0.23	0.01		
	ChatGPT	0.17	0.01	0.06	**0.01**
	MagicSchool.ai	0.20	0.01	0.03	**0.01**
Lemma Overlap in Adjacent Paragraphs (Text Cohesion)	Original	0.33	0.01		
	ChatGPT	0.25	0.01	0.08	**0.01**
	MagicSchool.ai	0.31	0.01	0.02	0.82
Dale-Chall Readability (Readability Formula)	Original	9.76	0.21		
	ChatGPT	8.48	0.22	1.28	**0.01**
	MagicSchool.ai	8.66	0.18	1.10	**0.01**
Flesch-Kincaid Grade Level (Readability Formula)	Original	9.50	0.35		
	ChatGPT	8.49	0.36	1.01	0.25
	MagicSchool.ai	8.26	0.29	1.24	**0.02**

Note. The EMMs are evaluated at Word Count = 389.82. **Bolded** *p*-values are significant

the post-hoc tests for the eight significant models (i.e., not including the models for All Connectives and Causal Connectives).

The results of the post-hoc tests supported Hypotheses 2.1 and 2.3. In other words, the altered texts used less sophisticated words than the original texts as measured by Age of Acquisition (altered texts had lower AoA), Word Concreteness (altered texts had higher concreteness), and Lexical Decision Response Time (altered texts had faster response time). In addition, the altered texts had lower grade-level readability as measured by the two traditional readability formulae. However, Hypothesis 2.2 was not supported, and in fact the opposite pattern was observed. The altered texts had less cohesion compared to the original texts.

Experiment 2 Discussion

The goal of Experiment 2 was to extend the design and procedure of Experiment 1 by using a different prompt for ChatGPT and testing a second LLM-based tool. The same 30 texts used in Experiment 1 were used in Experiment 2. The texts were altered by both ChatGPT4o and MagicSchool Text Leveler to be readable for a 6th grade student. It was hypothesized that compared to the original texts, the altered texts would use less sophisticated words, be more cohesive, and be scored as easier using traditional readability scores. The results showed that the altered texts used less sophisticated words and were easier according to traditional readability formulas. However, the altered texts were less cohesive than the original texts and neither LLM tool lowered the FKGL to the appropriate level. That is, when the LLMs were prompted to alter a text to 6th grade reading level, on average they rewrote the text to a 9th grade reading level. These limitations of the LLM tools demonstrate that further work is needed to effectively use LLMs to ease text readability.

General Discussion

The ability of generative LLMs to ease text readability was investigated in two experiments. Five prompts were tested across two versions of ChatGPT (4 and 4o) and a LLM tool specifically designed for altering text readability (MagicSchool.ai Text Leveler). The texts produced by LLMs were evaluated using NLP tools to measure linguistic sophistication, text cohesion, and traditional readability formulas. The results demonstrated that LLMs may be useful in easing text readability but likely require detailed prompts. In particular, no tool or prompt tested in this study effectively increased text cohesion. Altering text cohesion is necessary for improving readability, as readers of different skill levels better comprehend texts at different cohesion levels (McNamara & Kintsch, 1996).

The current study had three major limitations. First was the inclusion of only informational texts. While reading informational texts is a critical component of reading skills (NAEP, 2024), past research has found linguistic and syntactic differences between informational and literary texts (Sheehan et al., 2014). Thus, the study is not widely generalizable to other text genres and future research should extend the experiments of this paper to other text genres. Second, the sample size of 30 original texts may not have contained enough linguistic variety to afford LLMs the opportunity to simplify texts. Further research should be conducted to extend the methodology to larger corpora of texts. Additionally, recent research by Huynh and McNamara (2025) and (Potter et al., 2026) on different and larger text corpora found that generative LLMs had difficulty adapting the cohesion of texts. Finally, while

LLMs may struggle to adapt text readability, humans may not be particularly suited to the task either (Jensen et al., 2024). Thus, future research and development should focus on the comparison between human and AI strategies to ease text readability, and opportunities for human-AI collaboration on text readability tasks.

In addition to these major limitations, one potential confound for the linguistic measurements based on the Brysbaert Concreteness and Kuperman AoA corpora was that the original and altered texts had a relatively low proportion of words occurring in those corpora. On average, 95% or more of the words in all the texts occurred in the SUBTLEX and ELP corpora. In contrast, approximately 80% of the words in all the texts occurred in the Brysbaert corpora, and approximately 70% of the words in all the texts occurred in the AoA corpora.

Despite these limitations, there is a clear implication of these findings for educational practice: the use of generative LLMs for altering text readability is inadvisable without careful testing of the inputs and outputs. One clear issue found in the results of this paper is that while the tools used in Experiment 2 did ease text readability, the altered texts were still several grade levels away from the target grade level. Instructors without experience in evaluating LLM outputs may see texts with slightly eased readability as, "good enough", even though students reading those texts will struggle to comprehend the texts. As Imundo and colleagues (2024) warn, while experts may be better able than novices to evaluate information produced by generative AI, experts are still susceptible to miscalibrations and illusions generated by generative AI. Instructors should be careful when using generative AI to alter text readability as the outputs may be inappropriate for the intended students.

One unexpected finding of these experiments was that ChatGPT had a *brevity bias*. Across both experiments, the texts generated by ChatGPT consistently had fewer words compared to the original texts. In fact, for three of the four prompts, not a single altered text had more words than the original text (Prompts 1.1, 1.2, and 2.1: see Figs. 1 and 2). The brevity bias may further reduce a text's holistic readability, as texts with fewer words are typically less cohesive because fewer words are repeated across sentences and paragraphs. Recent research has found that ChatGPT produces sentence paraphrases comparable to the length of the original sentence (Kurt Pehlivanoğlu et al., 2024). However, to the knowledge of the authors there has not been a report on whether ChatGPT maintains text length when rewriting a text. The unexpected nature of this finding requires further research to confirm the presence of the brevity bias.

Overall, while LLMs show some promise in altering text readability, the results of this study demonstrate that more sophisticated techniques, and potentially tools, will be needed to effectively ease text readability. Indeed, one concerning interpretation of these results is that LLM-generated texts may appear easier at the surface level (i.e., the texts are shorter and use less sophisticated words), but the texts may be harder for students to understand because they are less cohesive. However, the major limitation of this study is that human participant data was not collected and indeed has typically not been collected in the initial studies on LLMs and text readability (e.g., Imperial & Madabushi, 2023 & Naseem, 2024 used NLP techniques to evaluate readability). Further studies with human subjects are necessary to compare the differences in readability between human-generated and LLM-generated texts.

Acknowledgements Thanks to Maria Goldshtein and Tracy Arner for providing feedback on the final version of the manuscript.

Author Contributions M.W. designed the experiment, collected and analyzed the data, and wrote the manuscript.

Funding The author declares that no funds, grants, or other support were received during the preparation of this manuscript.

Data Availability The text corpus and data for both experiments are available for peer-review at the open science framework via an anonymized link. The anonymized link will be updated to the public DOI upon acceptance of this paper for publication.

Declarations

Competing Interests The authors declare no competing interests.

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U.S. Department of Education Institute of Education Sciences, National Center for Education Statistics, National Assessment of Educational Progress (NAEP), 2024 Reading Assessment.